

13. PROBABILITATE BANAKETAK

1

415 | 1.

	x_i	f_i	$f_i x_i$	$f_i x_i^2$
(0,5]	2,5	2	5	12,5
(5,10]	7,5	11	82,5	618,75
(10,15]	12,5	13	162,5	2031,25
(15,20]	17,5	6	105	1837,5
(20,25]	22,5	3	67,5	1518,75
(25,30]	27,5	1	27,5	756,25
		36	450	6775

kalkulyatorok

$$\bar{x} = 12,5$$

$$\sum x = 450$$

$$\sum x^2 = 6775$$

$$\text{VAR } \sigma^2 x = 31,944$$

$$\text{DES. } \sigma x = 5,651$$

$$(\sigma^2 x = 32,857)$$

BATAZBESZREKA

$$\bar{x} = \frac{\sum f_i x_i}{N} = \frac{450}{36} = 12,5$$

BARJANTRA

$$\sigma^2 = \frac{\sum f_i x_i^2}{N} - \bar{x}^2 = \frac{6775}{36} - 12,5^2 = 31,944$$

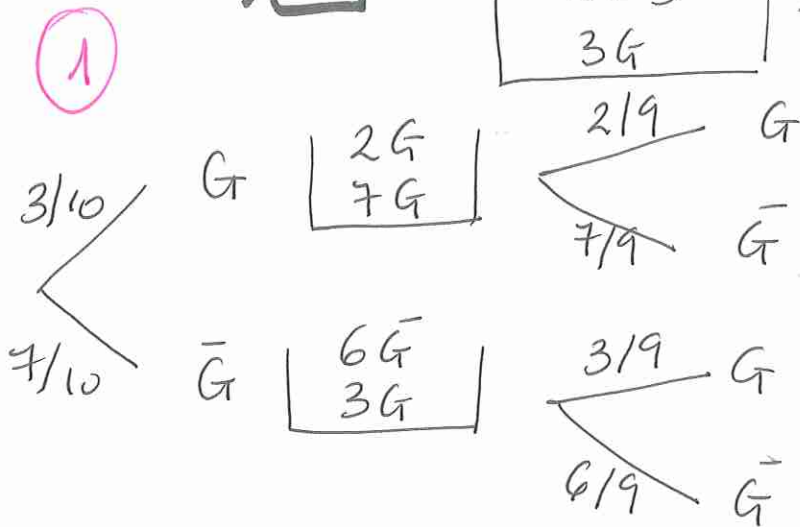
DESZIDERAPEN PRILKA

$$\sigma = \sqrt{\frac{\sum f_i x_i^2}{N} - \bar{x}^2} = \sqrt{31,944} = 5,65$$

$$\sigma = \sqrt{\frac{\sum (x_i - \bar{x})^2}{N}}$$

434 [3.]

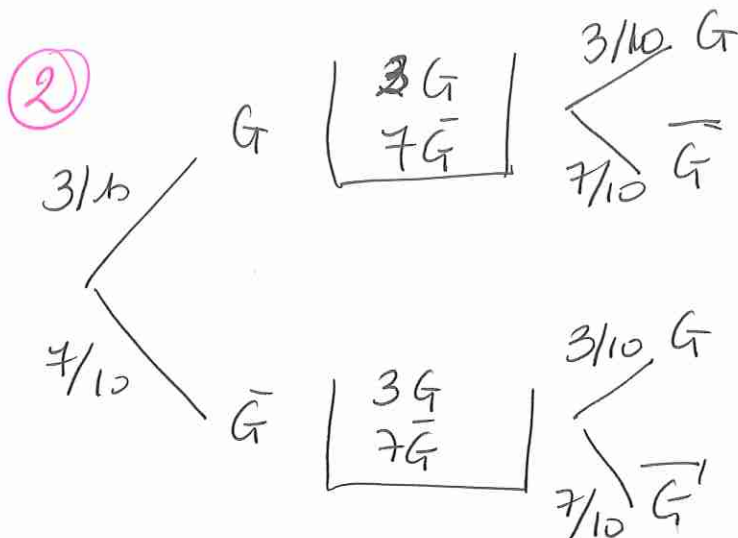
$$\left[\begin{array}{c} 5\bar{G} \\ 2B \\ 3G \end{array} \right] \Rightarrow \left[\begin{array}{c} 3G \\ 7\bar{G} \end{array} \right]$$



$$P(0) = \frac{7}{10} \cdot \frac{6}{9} = \frac{7}{15}$$

$$P(1) = 1 - \left(\frac{7}{15} + \frac{1}{15} \right) = \frac{7}{15}$$

$$P(2) = \frac{3}{10} \cdot \frac{2}{9} = \frac{6}{90} = \frac{1}{15}$$



$$P(0) = \frac{7}{10} \cdot \frac{7}{10} = \frac{49}{100}$$

$$P(1) = 1 - \left(\frac{49}{100} + \frac{9}{100} \right) = \frac{21}{50}$$

$$P(2) = \frac{3}{10} \cdot \frac{3}{10} = \frac{9}{100}$$

①

X_i	p_i
0	$7/15$
1	$7/15$
2	$1/15$
	1

②

X_i	p_i
0	$4/100$
1	$21/50 = 42/100$
2	$9/100$
	1

②. Kasus. Bawktu binomial da. $p = 3/10$ eto $n = 2$
 $B(2, 3/10)$

21

$$P(G) = 0.4$$

$$n = 5$$

$$p = 0.4 \quad q = 0.6$$

$$P(X=k) = \binom{n}{k} \cdot p^k \cdot q^{n-k}$$

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

SHIFT
nCr

Saiokuntza dikotomikoa da, erabaketa izatea $p = 0.4$ da eta erabaketa ez izatea $q = 0.6$.

Probabilitateak konstanteak dira, frogo bakiak erabaketa, beste aukerako frogakiko independenteak.
 $n = 5$ saiokuntza kopurua da.

BANAKETA BINOMIALA $\Rightarrow B(5; 0.4)$

$$P(X=0) = 0.6^5 = 0.08$$

$$P(X=1) = \binom{5}{1} \cdot 0.4^1 \cdot 0.6^4 = 0.26$$

$$P(X=2) = \binom{5}{2} \cdot 0.4^2 \cdot 0.6^3 = 0.34$$

$$P(X=3) = \binom{5}{3} \cdot 0.4^3 \cdot 0.6^2 = 0.23$$

$$P(X=4) = \binom{5}{4} \cdot 0.4^4 \cdot 0.6^1 = 0.08$$

$$P(X=5) = \binom{5}{5} \cdot 0.4^5 \cdot 0.6^0 = 0.01$$

x_i	p_i	$p_i x_i$	$p_i x_i^2$
0	0.08	0	0
1	0.26	0.26	0.26
2	0.34	0.68	1.36
3	0.23	0.69	2.07
4	0.08	0.32	1.28
5	0.01	0.05	0.25
<u>1</u>		<u>2</u>	<u>5.22</u>

$$\mu = \sum p_i x_i = 2$$

$$\sigma = \sqrt{\sum p_i x_i^2 - \mu^2} =$$

$$= \sqrt{5.22 - 2^2} = \underline{\underline{1.1}}$$

$$\mu = 2 \quad \text{BARRA BEST}$$

$$\sigma = 1.1 \quad \text{DEBIDERAKO TIPIKOA}$$

417) (2)

DADO BAT LAURİ ETİ

PONTAŞIARI DAĞIKION PROBABILITY -
BANAİKTAREN μ eta σ .

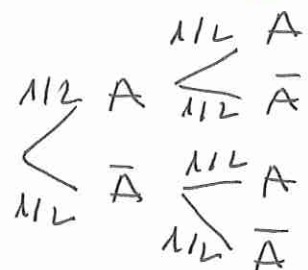
x_i	p_i	$p_i x_i$	$p_i x_i^2$
1	1/6	1/6	1/6
2	1/6	2/6	4/6
3	1/6	3/6	9/6
4	1/6	4/6	16/6
5	1/6	5/6	25/6
6	1/6	1	6
	1	21/6	91/6

$$\mu = \sum p_i x_i = 21/6$$

$$\sigma = \sqrt{\sum p_i x_i^2 - \mu^2} =$$

$$\sqrt{\frac{91}{6} - \left(\frac{21}{6}\right)^2} = 1.71$$

(3)

Bİ TRANDON BOTAREN BADIYU, 0,1 eta 2
AURPEİİ WR DITAKU.KALKULATU PROBABILITY - BANAİKTAREN
BATELRESİKİA ETİ DESBİATİ TİKİA.

$$P(0) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

$$P(1) = 1 - \left(\frac{1}{4} + \frac{1}{4}\right) = \frac{2}{4} = \frac{1}{2}$$

$$P(2) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

x_i	p_i	$x_i p_i$	$p_i x_i^2$
0	1/4	0	0
1	1/2	1/2	1/2
2	1/4	2/4	1
	1	1	6/4

$$\mu = \sum x_i p_i = 1$$

$$\sigma = \sqrt{\sum p_i x_i^2 - \mu^2}$$

$$= \sqrt{6/4 - 1^2} =$$

$$= 0.71$$

434) (4) Berferi BANAKETA BINOMIALA.

$$\mu = np$$

$$\sigma = \sqrt{npq}$$

a) $n = 50$. $p = 1/3 \rightarrow B(50, 1/3)$.

$1/3$ $\bar{z}$
 $2/3$ $\bar{z}$

$q = 2/3$.

$$\mu = np = 50 \cdot \frac{1}{3} = 50/3 = \underline{\underline{16,67}}$$

$$\sigma = \sqrt{50 \cdot \frac{1}{3} \cdot \frac{2}{3}} = \underline{\underline{3,33}}$$

b)

c) $n = 400$ $p = 1/2$ $q = 1/2 \rightarrow B(400, 1/2)$

$1/2$ A
 $1/2$ $\bar{A}$

$$\mu = np = 400 \cdot 1/2 = \underline{\underline{200}}$$

$$\sigma = \sqrt{npq} = \sqrt{400 \cdot 1/2 \cdot 1/2} = \underline{\underline{10}}$$

d) $n = 1000$
 $p = 0,01$
 $q = 0,99$

$$B(1000; 0,01)$$

$$\mu = np = 1000 \cdot 0,01 = \underline{\underline{10}}$$

$$\sigma = \sqrt{npq} = \sqrt{1000 \cdot 0,01 \cdot 0,99} = \underline{\underline{3,15}}$$

434) (5) $B(9; 0,2)$

$$P[X=k] = \binom{n}{k} p^k q^{n-k}$$

$$\binom{n}{p} = \frac{n!}{p!(n-p)!}$$

a) $P[X < 3] = P(X=0) + P(X=1) + P(X=2)$

$$P[X=0] = \binom{9}{0} 0,2^0 \cdot 0,8^9 = \dots \quad 0,8^9 = 0,134$$

$$P[X=1] = \binom{9}{1} 0,2^1 \cdot 0,8^8 = \frac{9!}{1!8!} 0,2^1 \cdot 0,8^8 = 0,302$$

$$P[X=2] = \binom{9}{2} 0,2^2 \cdot 0,8^7 = \frac{9!}{2!7!} 0,2^2 \cdot 0,8^7 = 0,302$$

$$P[X < 3] = 0,134 + 0,302 + 0,302 = \underline{\underline{0,738}}$$

$$b) P[X \geq 7] = P[X=7] + P[X=8] + P[X=9]$$

$$P[X=7] = \binom{9}{7} 0,2^7 \cdot 0,8^2 = 0,0003$$

$$P[X=8] = \binom{9}{8} 0,2^8 \cdot 0,8^1 = 0,00002$$

$$P[X=9] = \binom{9}{9} 0,2^9 \cdot 0,8^0 = 0,2^9 = 0,0000005$$

$$P[X \geq 7] = 0,0003 + 0,00002 + 0,0000005 = \underline{\underline{0,0003205}}$$

$$c) P[X \neq 0] = 1 - P[X=0] = 1 - 0,134 = \underline{\underline{0,8658}}$$

$$d) P[X \leq 9] = 1$$

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2 AKASTINAK.

50 torlojuko koxatan

ze prob. daqoen koxa batean torloju akastunuen kopurua hau izateko: $\Rightarrow B(50; 0,02)$

a) Bat ere et

$$P[X=0] = 0,98^{50} = \underline{\underline{0,36}}$$

b) Bat

$$P[X=1] = \binom{50}{1} 0,02^1 \cdot 0,98^{49} = \underline{\underline{0,37}}$$

c) Bi baino gehiago.

$$P[X > 2] = 1 - P[X \leq 2] = 1 - [P[X=0] + P[X=1] + P[X=2]]$$

$$= 1 - [0,36 + 0,37 + 0,19] = \underline{\underline{0,08}}$$

$$P[X=2] = \binom{50}{2} 0,02^2 \cdot 0,98^{48} = 0,19$$

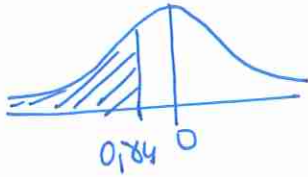
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BANAKETA NORVA LA

1

N(0,1)

$$P[z \leq 0,84] = 0,7996$$



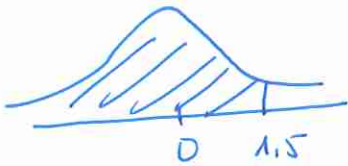
$$e) P[z < 2,35] = 0,9906$$

$$f) P[z \leq 0] = 0,5$$

$$g) P[z < 4] = 1$$



$$b) P[z < 1,5] = 0,9332$$

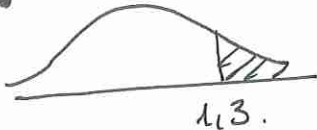


$$h) P[z = 1] = 0$$

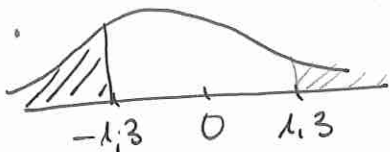
$$c) P[z < 2] = 0,9772$$

$$d) P[z < 1,87] = 0,9693$$

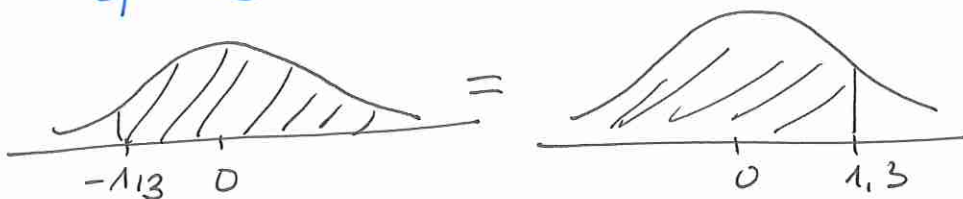
$$\Rightarrow 4) P[z > 1,3] = 1 - P(z \leq 1,3) = 1 - 0,9032 = \underline{0,0968}$$



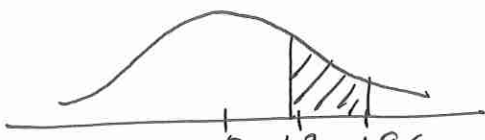
$$b) P[z < -1,3] = P(z \geq 1,3) = 1 - P(z \leq 1,3) = 1 - 0,9032 = \underline{0,0968}$$



$$c) P[z > -1,3] = P[z \leq 1,3] = \underline{0,9032}$$



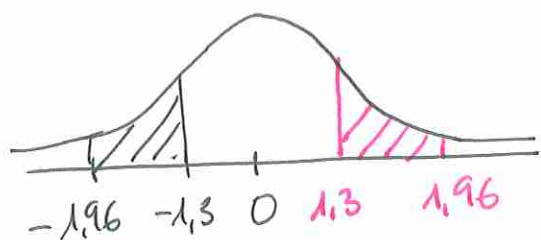
$$d) P[1,3 < z < 1,96] = P[z \leq 1,96] - P[z \leq 1,3] = 0,9750 - 0,9032 = \underline{0,0718}$$



$$e) P[-1,96 < z < -1,3] = P[1,3 < z < 1,96] =$$

$$= P[z \leq 1,96] - P[z \leq 1,3] =$$

$$= 0,9750 - 0,9032 = \underline{\underline{0,0718}}$$



$$f) P[-1,3 < z < 1,96] = P[z \leq 1,96] - P[z \leq -1,3] =$$

$$= P[z \leq 1,96] - P[z \geq 1,3] =$$

$$= P[z \leq 1,96] - [1 - P[z \leq 1,3]] =$$

$$= P[z \leq 1,96] - 1 + P[z \leq 1,3] =$$

$$= 0,9750 - 1 + 0,9032 = \underline{\underline{0,8782}}$$

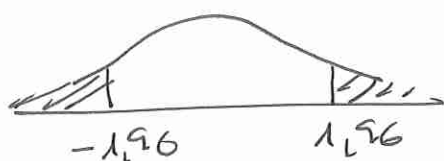
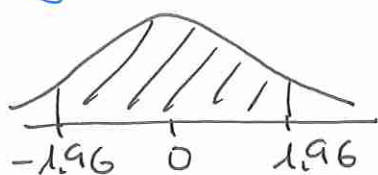


$$g) P[-1,96 < z < 1,96] = P[z \leq 1,96] - P[z \leq -1,96] =$$

$$= P[z \leq 1,96] - P[z \geq 1,96] =$$

$$= P[z \leq 1,96] - [1 - P[z \leq 1,96]] =$$

$$= 0,9750 - 1 + 0,9750 = \underline{\underline{0,95}}$$



434) **8** a) $P[z=2] = 0.$

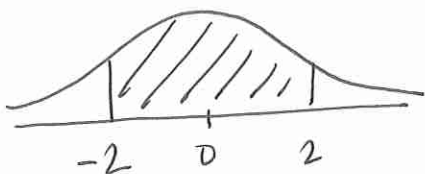
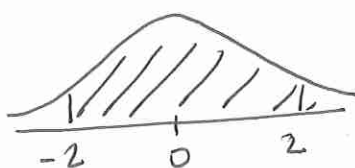
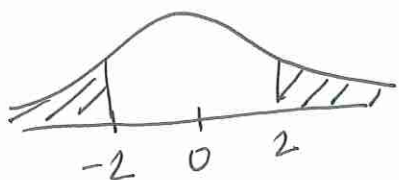
b) $P[z \leq 2] = 0,9772$

c) $P[z \geq 2] = 1 - P[z \leq 2] = 1 - 0,9772 = 0,0228$

d) $P[z \leq -2] = P[z \geq 2] = 1 - P[z \leq 2] = 1 - 0,9772 = 0,0228$

e) $P[z > -2] = P[z \leq 2] = 0,9772$

f) $P[-2 \leq z \leq 2] = P[z \leq 2] - P[z \leq -2] = 0,9772 - 0,0228 = 0,9544$



9-10-11.

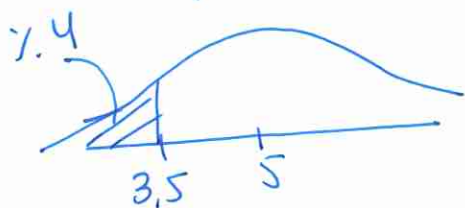
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BANAKETA NORMALIS

A metode

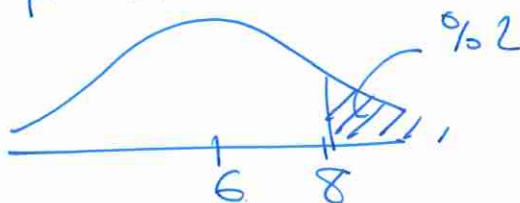
$\mu = 5$



lehenengo kalkulatu

B metode

$\mu = 6$



bi banaketa normalen desbiderazpen tipikoa behar da.

A METODOAN:

$P[X \leq 3,5] = 0,04.$

$P\left[z \leq \frac{3,5 - 5}{\sigma_A}\right] = 0,04$

$P\left[z \leq -\frac{1,5}{\sigma_A}\right] = P\left[z \geq \frac{1,5}{\sigma_A}\right] = 0,04$

$P\left[z \leq \frac{1,5}{\sigma_A}\right] = 1 - 0,04 = 0,96$

1.7 → 0.96
0.9599

Taulahk 1.75 = $\frac{1,5}{\sigma_A} \rightarrow \sigma_A = 0,86$

B. METODOJAN:

$$X_2 \sim N(6, \sigma_2)$$

$$P[X_2 \geq 8] = 0.02$$

$$P\left[Z \geq \frac{8-6}{\sigma_2}\right] = 0.02 \Rightarrow P\left[Z \geq \frac{2}{\sigma_2}\right] = 0.02$$

Taulatik

$$\begin{array}{c} \uparrow 5 \\ 2 \leftarrow \boxed{0.98} \end{array}$$

$$P\left[Z \leq \frac{2}{\sigma_2}\right] = 1 - 0.02 = \boxed{0.98}$$

$$\underline{\underline{2.05}} \rightarrow \frac{2}{\sigma_2} = 2.05 \rightarrow \underline{\underline{\sigma_2 = 0.98}}$$

A metodoo

$$X \sim N(5, 0.986)$$

B metodoo

$$X \sim N(6, 0.98)$$

a) A metodorekiu zer eluruekok ez du lotuko 6,5 ek
ustoa (oinaditzae)?

$$Z = \frac{X - \mu}{\sigma}$$

$$P[X \leq 6.5] = P\left[Z \leq \frac{6.5 - 5}{0.986}\right] = P[Z \leq 1.74] = 0.9591$$

$$\rightarrow \approx \boxed{7.96\%}$$

b) B metodorekiu zer eluruekok lotuko du 4 eta 6 artekoak.

$$P[4 \leq X \leq 6] = P\left[\frac{4-6}{0.98} \leq Z \leq \frac{6-6}{0.98}\right] = P[-2.04 \leq Z \leq 0]$$



$$\begin{aligned} &= P[0 \leq Z \leq 2.04] = P[Z \leq 2.04] - P[Z \leq 0] = \\ &= 0.9793 - 0.5 = \boxed{0.4793} \Rightarrow \underline{\underline{48\%}} \end{aligned}$$

31Banaketa normala. $\Rightarrow X \sim N(\underbrace{1500}_{\mu}, \underbrace{200}_{\sigma})$.

$$\mu = 1500 \text{ h}$$

$$\sigma = 200 \text{ h.}$$

a) zer prob dago zorian foku bat hartu eta 1000 urdu iraute gutxienez?

$$z = \frac{x - \mu}{\sigma}$$

$$P[X \geq 1000] = P\left[z \geq \frac{1000 - 1500}{200}\right] = P[z \geq -2.5] =$$

$$= P[z \leq 2.5] = \boxed{0.9938}$$



b) 2000 foku. badino prob!!

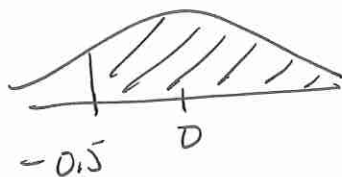
% 99.38 . 1000 urdu iraute gutxienez.

$$2000 \cdot \frac{99.38}{100} = 1987.6 \approx \underline{\underline{1987 \text{ foku}}}$$

32Banaketa normala. $X \sim N(\underbrace{75}_{\mu}, \underbrace{8}_{\sigma})$ $z = \frac{x - \mu}{\sigma}$

a) 71 kg tk gora.

$$P[X \geq 71] = P\left[z \geq \frac{71 - 75}{8}\right] = P[z \geq -0.5] = P[z \leq 0.5] = \underline{\underline{0.6915}}$$



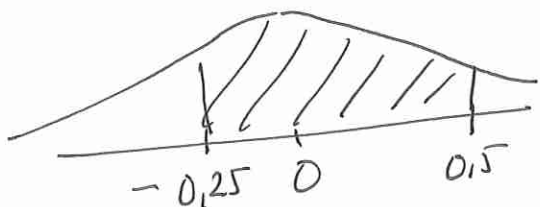
b) 73 eta 79 kg artean.

$$P[73 \leq X \leq 79] = P\left[\frac{73 - 75}{8} \leq z \leq \frac{79 - 75}{8}\right] =$$

$$= P[-0.25 \leq z \leq 0.5] = P[z \leq 0.5] - P[z \leq -0.25] =$$

$$= P[z \leq 0.5] - [1 - P(z \leq 0.25)] =$$

$$= 0.6915 - 1 + 0.5987 = \boxed{0.2902}$$

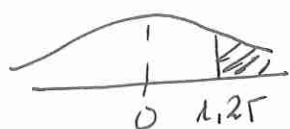


c) 80 kg-tik behera.

$$P[X \leq 80] = P\left[Z \leq \frac{80-75}{8}\right] = P[Z \leq 0,63] = \boxed{0,7357}$$

d) 85 kg-tik gora.

$$P[X \geq 85] = P\left[Z \geq \frac{85-75}{8}\right] = P[Z \geq 1,25] =$$



$$= 1 - P[Z \leq 1,25] = 1 - 0,8944 = \boxed{0,1056}$$

BINOMIALA $\rightarrow$ NORMALA

uz4.ori

16

Dado bat 1000 bider estatiku bodegu zer prob dago estatiku kopurua 100 baino txikiagoa izateko?

Banasketa BINOMIALA $B(n, p) \Rightarrow \boxed{X \sim B(1000, \frac{1}{6})}$

Banasketa binomiala normala bihurtzek baldintzak:

$$n \geq 10$$

$$n = 1000 \geq 10 \quad \checkmark$$

$$np \geq 5$$

$$np = 1000 \cdot \frac{1}{6} = 166,67 \geq 5 \quad \checkmark$$

$$np \geq 5$$

$$np = 1000 \cdot \frac{5}{6} \geq 5 \quad \checkmark$$

$$B(n, p) \rightarrow N(np, \sqrt{npq})$$

$$\sqrt{npq} = \sqrt{1000 \cdot \frac{1}{6} \cdot \frac{5}{6}} = 11,79$$

$$\rightarrow X \sim B(1000, \frac{1}{6}) \rightarrow X \sim N(166,67, 11,79)$$

Binomiala, normala bihurtzekoan matasen zuzenki eriten da

BINOMIALA

$$X \sim B(n, p)$$

$$X \sim B(1000, \frac{1}{6})$$

NORMALA

$$X' \sim N(np, \sqrt{npq})$$

$$X' \sim N(166,67; 11,79)$$

NORMAL TIPIFIKATIBA (14)

$$Z \sim N(0, 1)$$

$$P[X < 100] = P[X' < 99,5] = P\left[Z < \frac{99,5 - 166,67}{11,79}\right]$$

$$= P[Z \leq -5,7] = P[Z \geq 5,7] = 1 - P[Z \leq 5,7] \approx \underline{\underline{0}}$$

17

Txauru bat 400 bidu batz.

a) Prob: lotutako aurpegiu kopurua ≥ 200 .

• Txauru bat 400 bidu batzaren prob. jarraitzen du
BANAKETA BINOMIALA $B(400, \frac{1}{2})$

• Banaketa BINOMIALA, NORMALA izateko baldintzak:

$$n \geq 10 \quad 400 \geq 10 \quad \checkmark$$

$$np \geq 5 \quad 400 \cdot \frac{1}{2} = 200 \geq 5 \quad \checkmark$$

$$nq \geq 5 \quad 400 \cdot \frac{1}{2} = 200 \geq 5 \quad \checkmark$$

$$N(np, \sqrt{npq}) = N(200, 10)$$

$$\sqrt{400 \cdot \frac{1}{2} \cdot \frac{1}{2}} = 10$$

• Binomiala, normala bihurtzeko WATES-en zuzenketa eratu behar da.

BINOMIALA

$$X \sim B(400, \frac{1}{2})$$

NORMALA (yate)

$$X' \sim N(200, 10)$$

NORMAL TIPIFIKATIBA

$$Z \sim N(0, 1)$$

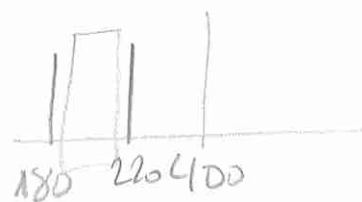
$$P[X > 200] = P[X' \geq 200,5] = P\left[Z \geq \frac{200,5 - 200}{10}\right] =$$



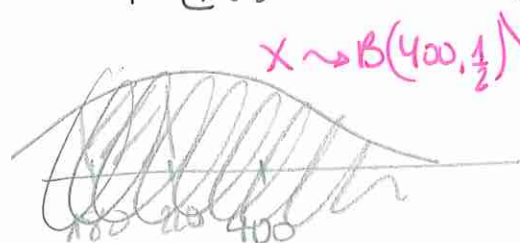
$$= P[Z \geq 0,05] = 1 - P[Z \leq 0,05] = 1 - 0,5199 =$$

$$= \boxed{0,4801}.$$

b) 180 eta 220 artean egoteko.



$$P[180 < x < 220] = P[180,5 \leq x' \leq 219,5]$$



$x \sim B(400, \frac{1}{2})$

yates

$x' \sim N(200, 10)$

$$= P\left[\frac{180,5 - 200}{10} \leq z \leq \frac{219,5 - 200}{10}\right] =$$

$z \sim N(0, 1)$

$$= P[-1,95 \leq z \leq 1] = P[z \leq 1] - P[z \leq -1,95] =$$



$$= P[z \leq 1] - [1 - P(z \leq 1,95)] =$$

$$= P[z \leq 1] - 1 + P[z \leq 1,95] =$$

$$?? = 0,8413 - 1 + 0,9744 = \boxed{0,8157}$$